Digital System Design

CAD Tools

TA Session 2

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Quartus II Basic Training

Quartus II Development System
Feature Overview

Software & Development Tools

- Quartus II
  - All Stratix, Cyclone & Hardcopy Devices
  - APEX II, APEX 20K/E/C, Excalibur, & Mercury Devices
  - FLEX 10K/A/E, ACEX 1K, FLEX 6000 Devices
  - MAX II, MAX 7000S/AE/B, MAX 3000A Devices
- Quartus II Web Edition
  - Free Version
  - Not All Features & Devices Included
    - See www.altera.com for Feature Comparison
- MAX+PLUS® II
  - All FLEX, ACEX, & MAX Devices

Quartus II Development System

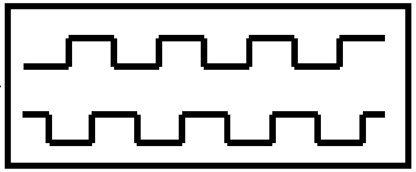
- Fully-Integrated Design Tool
  - Multiple Design Entry Methods
  - Logic Synthesis
  - Place & Route
  - Simulation
  - Timing & Power Analysis
  - Device Programming

Typical PLD Design Flow

Design Specification

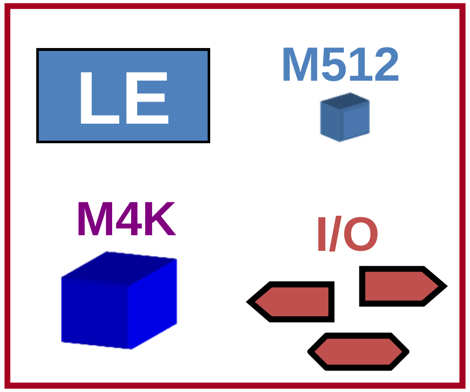
Design Entry/RTL Coding

- Behavioral or Structural Description of Design



RTL Simulation

- Functional Simulation (Modelsim®, Quartus II)
- Verify Logic Model & Data Flow (No Timing Delays)



Synthesis

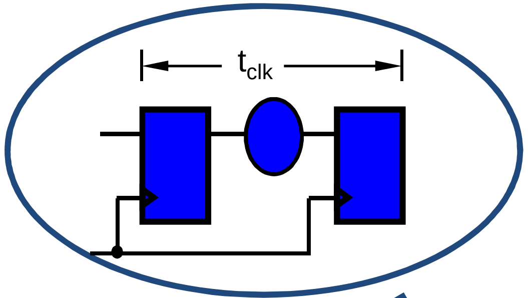
- Translate Design into Device Specific Primitives
- Optimization to Meet Required Area & Performance Constraints
- Precision Synthesis, Synplify/Synplify Pro, Design Compiler FPGA, Quartus II

QUARTUS® II

Place & Route

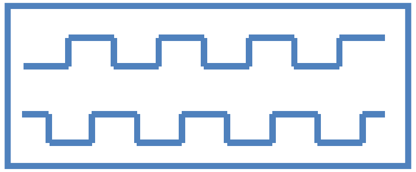
- Map Primitives to Specific Locations inside Target Technology with Reference to Area & Performance Constraints
- Specify Routing Resources to Be Used

Typical PLD Design Flow



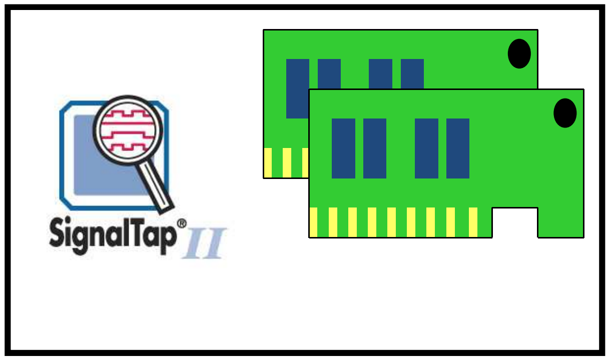
Timing Analysis

- Verify Performance Specifications Were Met
- Static Timing Analysis



Gate Level Simulation

- Timing Simulation
- Verify Design Will Work in Target Technology

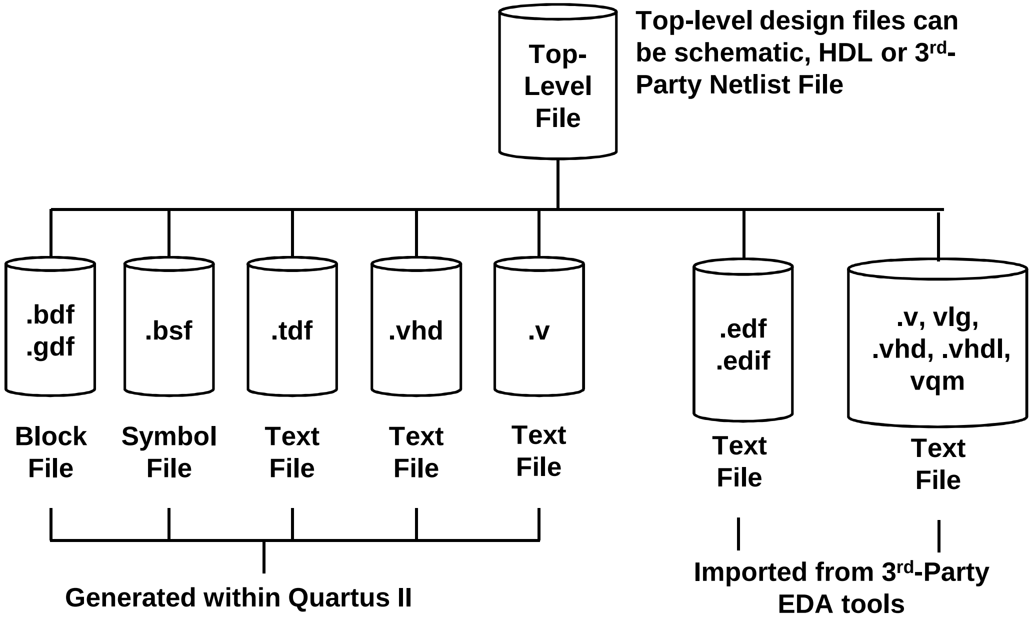


PC Board Simulation & Test

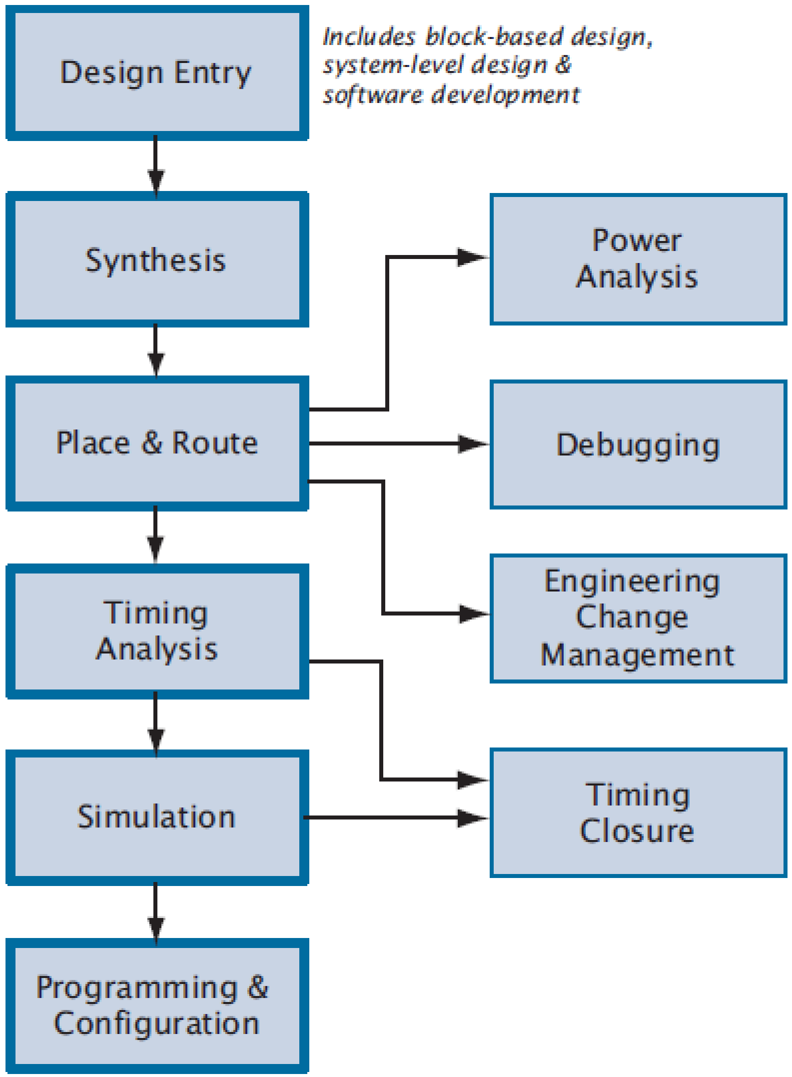
- Simulate Board Design
- Program & Test Device on Board
- Use **SignalTap II** for Debugging

Design Entry Methods

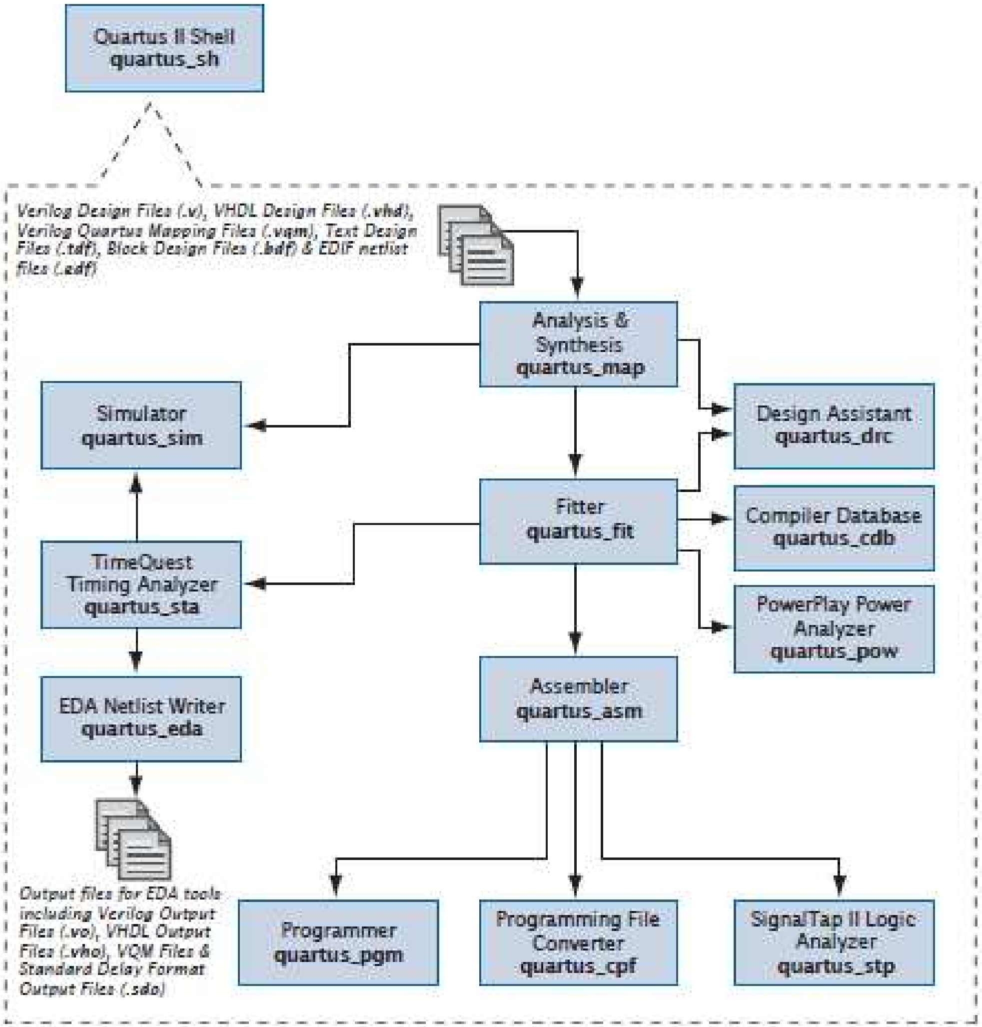
- Quartus II
  - Text Editor
    - AHDL
    - VHDL
    - Verilog
  - Schematic Editor
    - Block Diagram File
    - Graphic Design File
  - Memory Editor
    - HEX
    - MIF
- 3rd-Party EDA Tools
  - EDIF
  - HDL
  - VQM
- Mixing & Matching Design Files Allowed



Quartus II Design Flow



Command Line Design Flow



Quaruts GUI Features

Design Entry
- Text Editor
- Block & Symbol Editor
- MegaWizard Plug-In Manager
- Assignment Editor
- Floorplan Editor

Synthesis
- Analysis & Synthesis
- VHDL, Verilog HDL & AHDL
- Design Assistant
- RTL Viewer
- Technology Map Viewer

Timing Analysis
- Timing Analyzer
- Report Window
- Technology Map Viewer

Place & Route
- Fitter
- Assignment Editor
- Floorplan Editor
- Chip Editor
- Report Window
- Incremental Fitting
- Resource Optimization Advisor

Quartus II Basic Training

Quartus II Quick Start

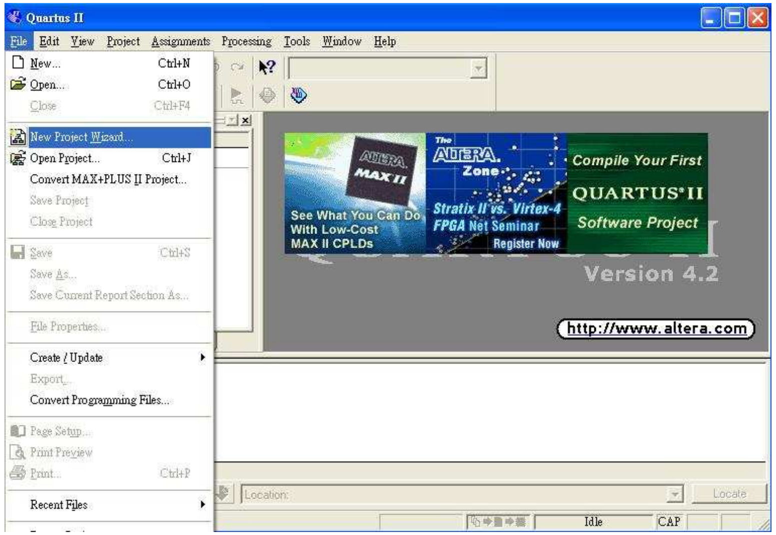
Part1

Objectives

- *Create a project using the New Project Wizard*
- *Name the project*
- *Add design files*
- *Pick a device*

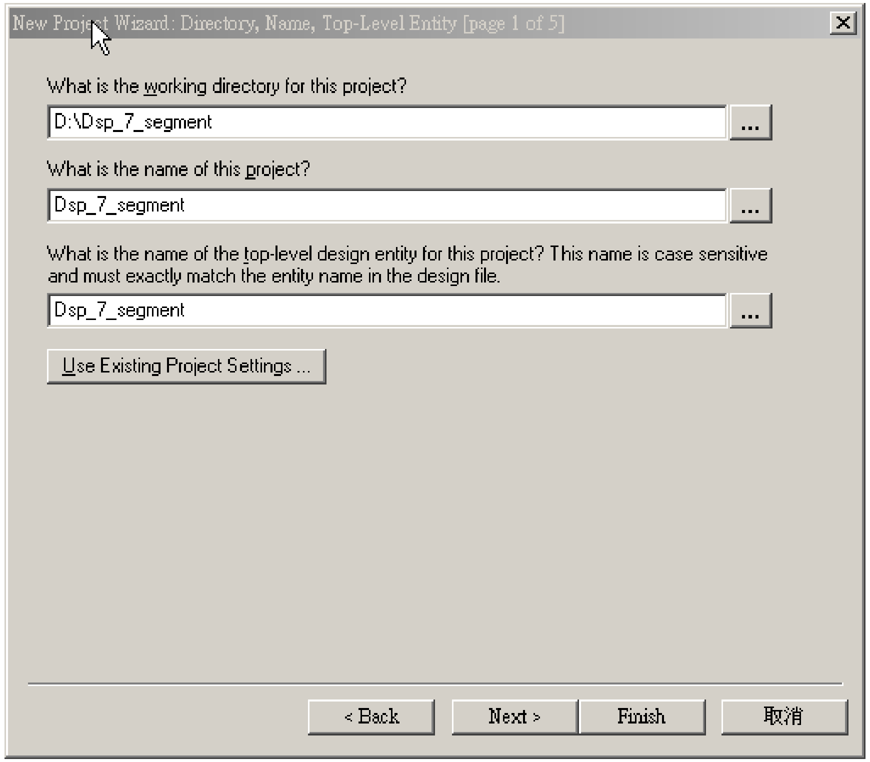
Step 1

Under File, Select **New Project Wizard**….
A new window appears. If an Introduction screen appears, click Next.



Step 2

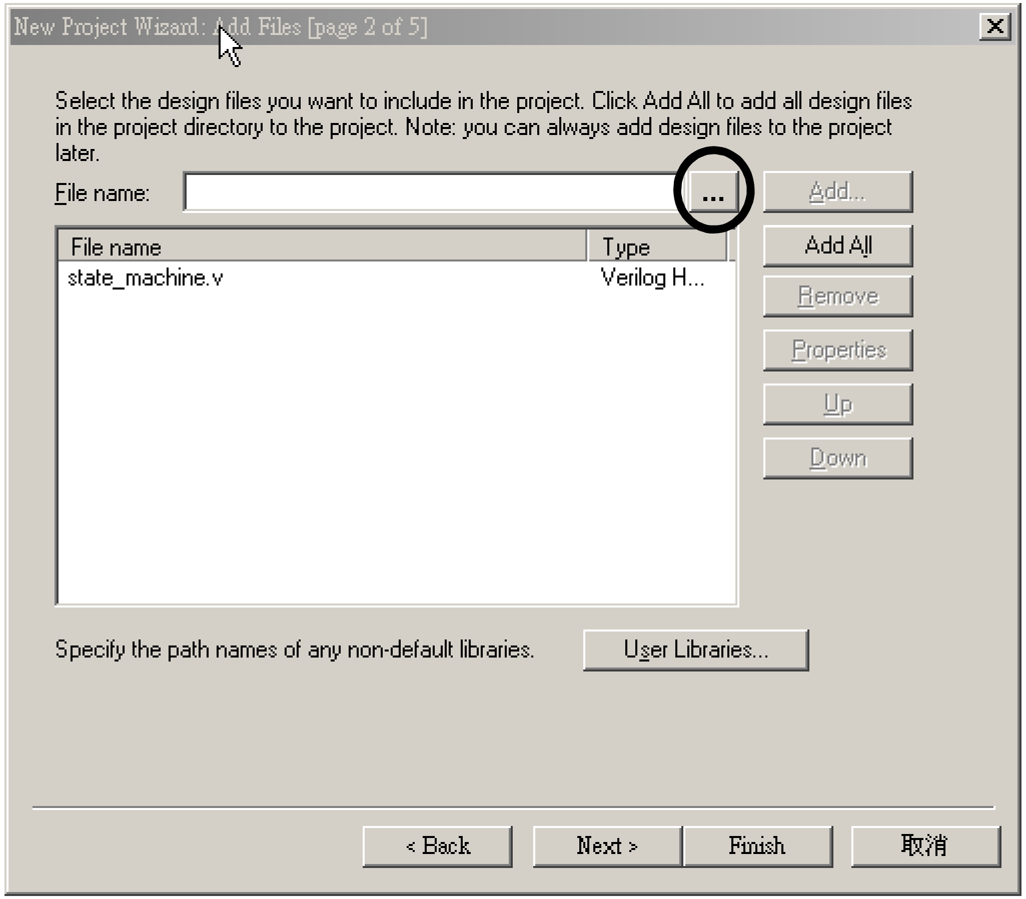
Page 1 of the wizard should be completed with the following



***Click Next** to advance to the*
Project Wizard: Add Files

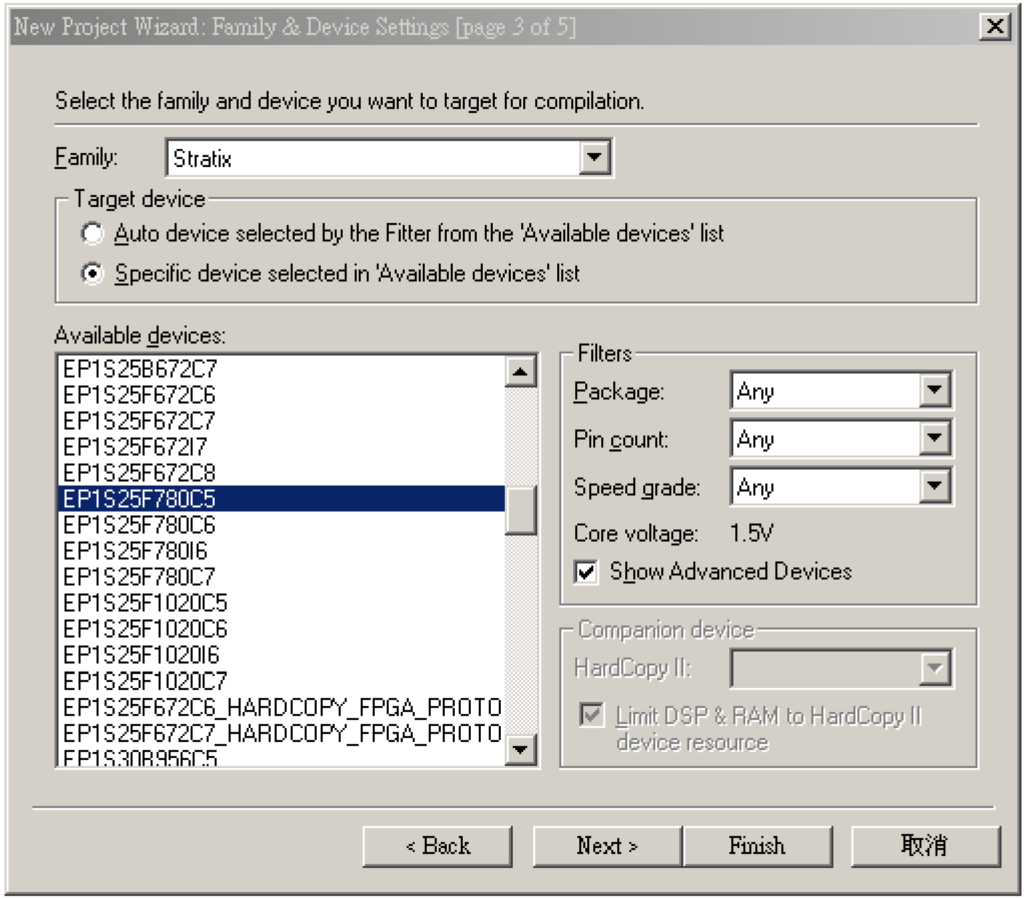
Step 3

Using the browse button, select **.v file**
Add to the project. Click **Next**.



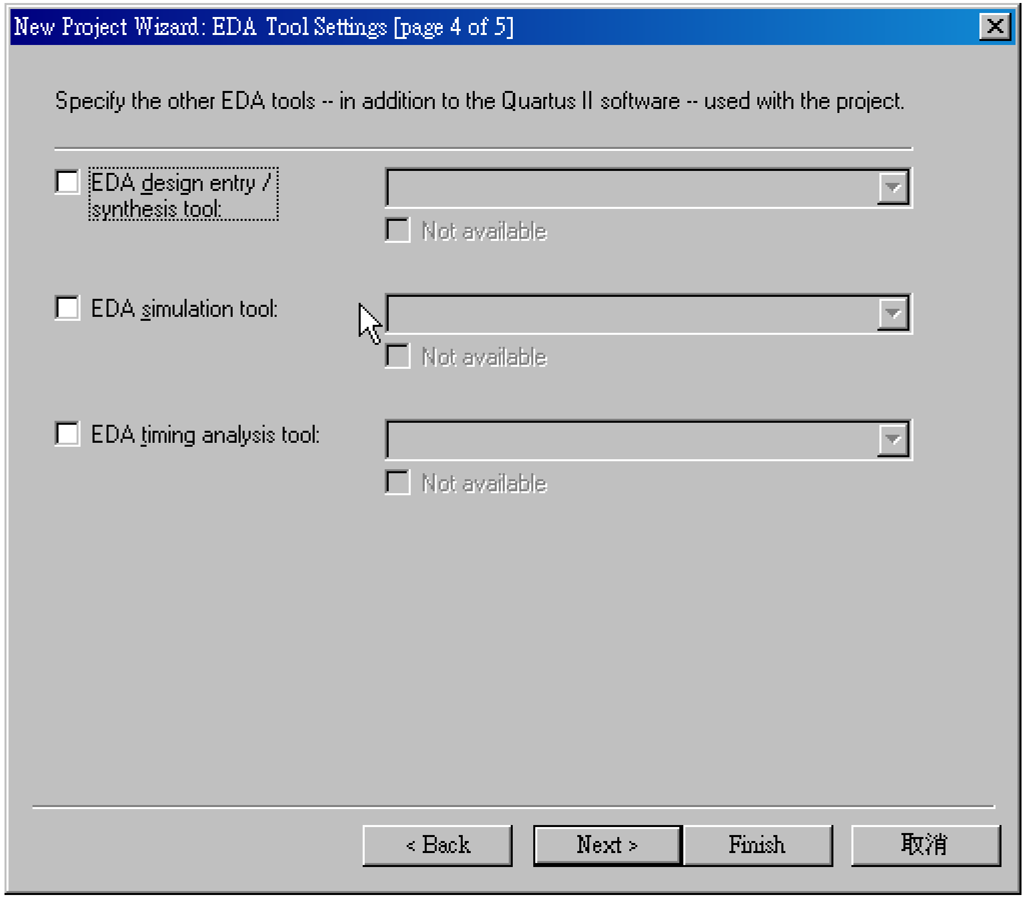
Step 4

On **page 3**, select the **Family**. Also, in the **Filters** section, set **Package** to **FBFA**, **Pin count** to **780**, and **Speed grade** to **5**. Select the device from the **Available devices:** window. Click **Next**.



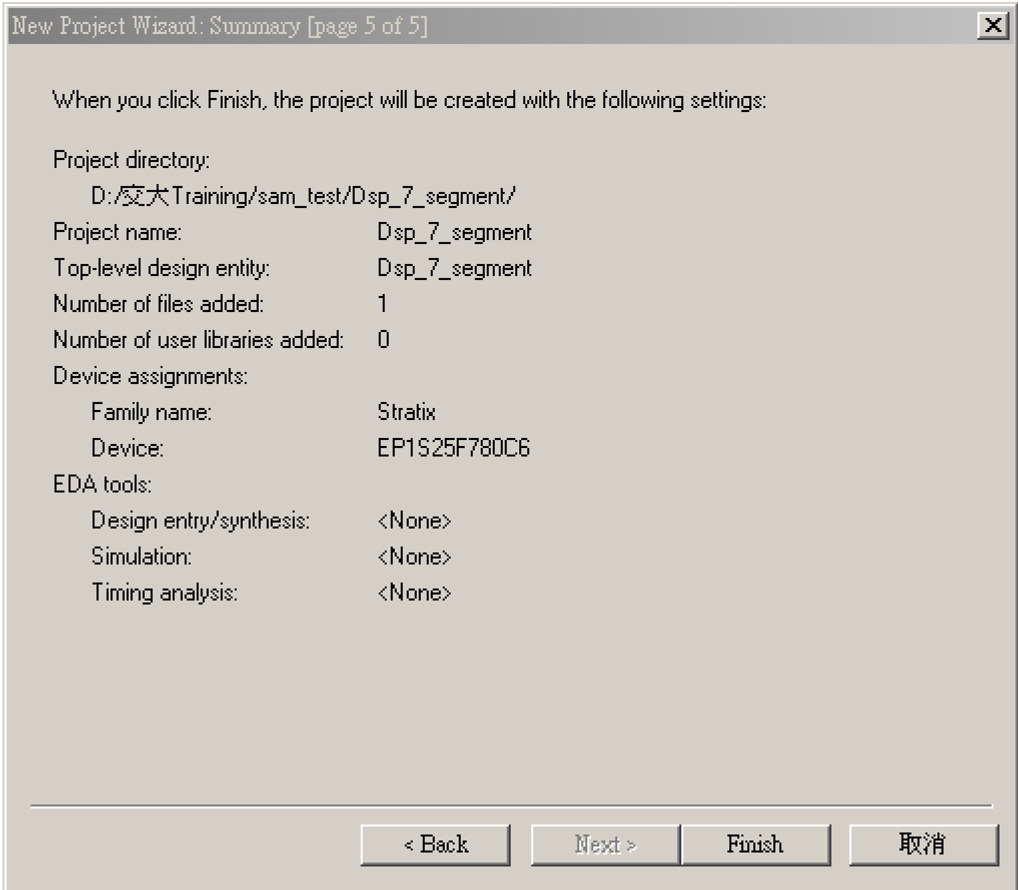
Step 5

On page 4 , you can specify any third party EDA tools you may be using along with Quartus II. Since these exercises will be done entirely within Quartus II, click **Next.**



Step 6

The summary screen appears as shown. Click **Finish.**
The project is now created.



Quartus II Basic Training

Quartus II Quick Start

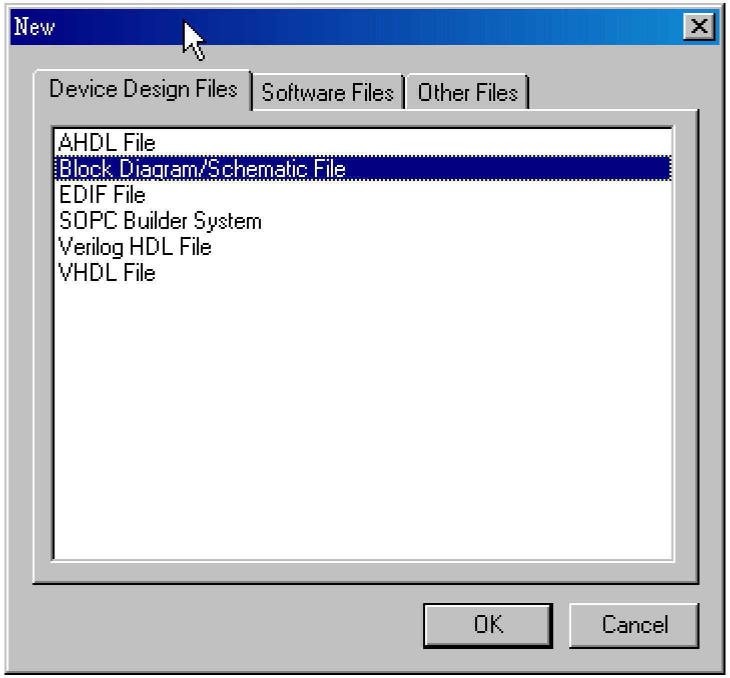
Part2

Objectives

- *Create a counter using the MegaWizard Plug-in Manager*
- *Build a design using the schematic editor*
- *Analyze and elaborate the design to check for errors*

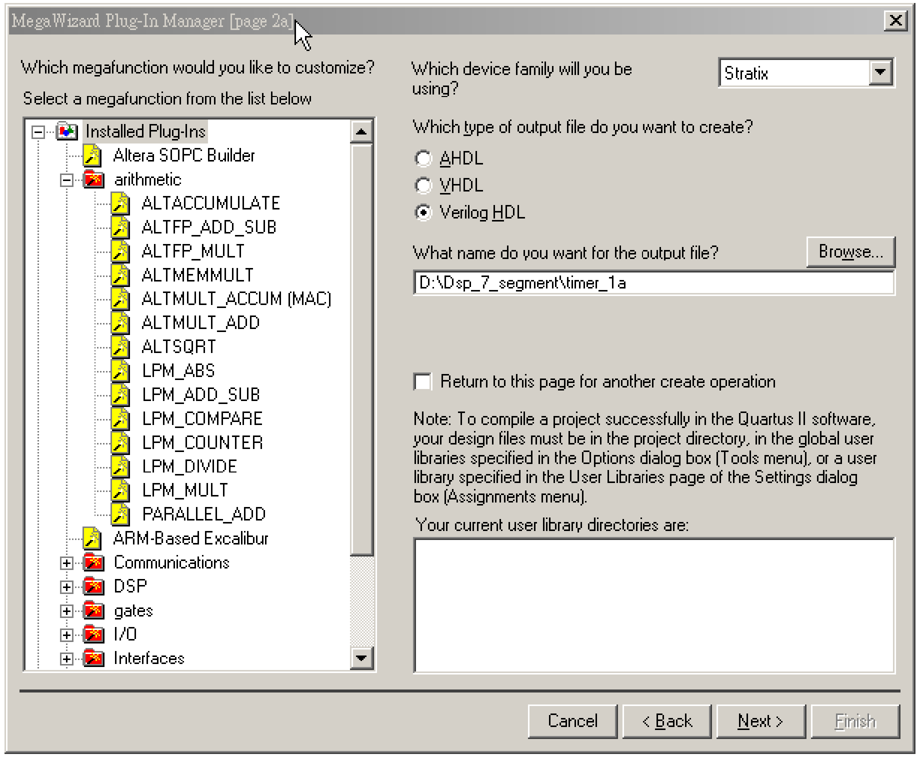
Step 1 Create schematic file

Select **File** ⇒ **New** and select **Block Diagram/Schematic File**. Click **OK**.
Select **File** ⇒ **Save As** and save the file as



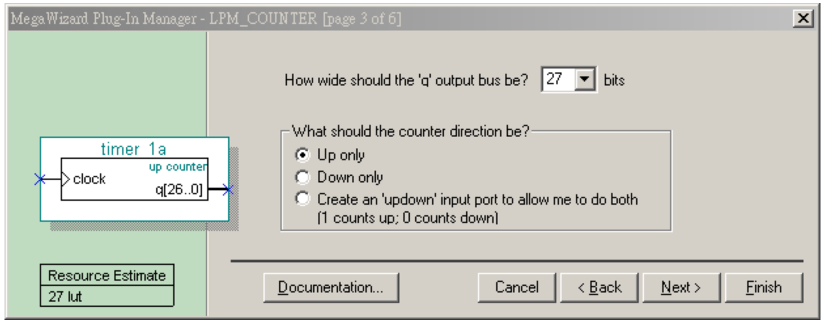
Step 2 Build an 23 bits counter using the MegaWizard Plug-in Manager

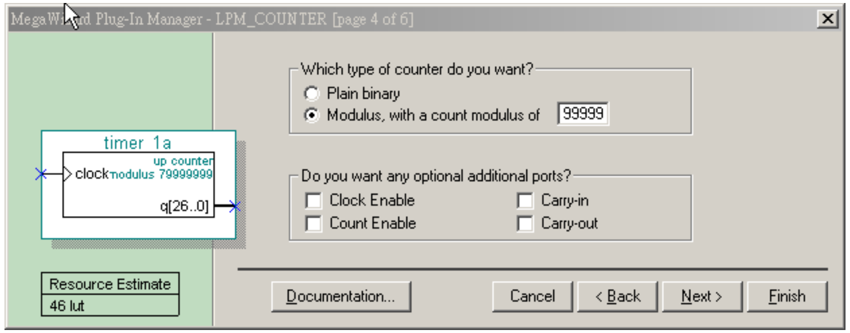
1. Choose **Tools** ⇒ **MegaWizard Plug-In Manager**. In the window that appears, select **Create a new custom megafunction variation**. Click on **Next**.
2. On **page 2a** of the **MegaWizard** expand the **arithmetic** folder and select **LPM_COUNTER**.
3. Choose **Verilog HDL** output For the name of the **output file,** type **timer_1s**. Click on **Next**

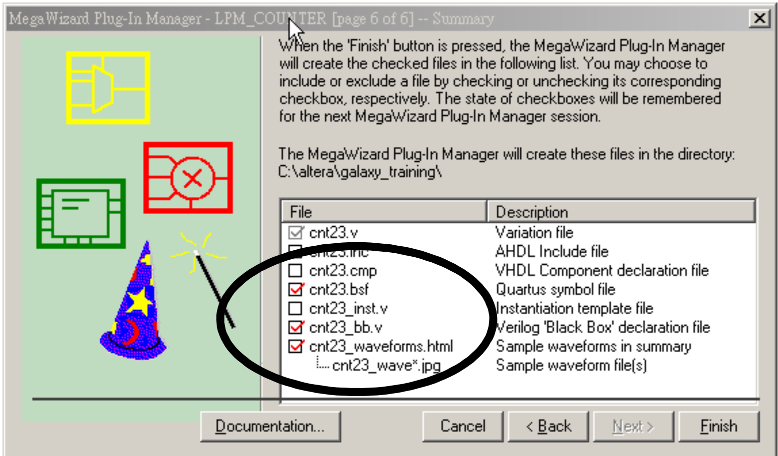


Step 3

1. Set the output bus to **27** bits. For the remaining settings in this window, use the defaults that appear .. Select **next**
2. .Turn on “**Modulus , with a count modulus of** “and key in **79999999**
3. Select **finish**

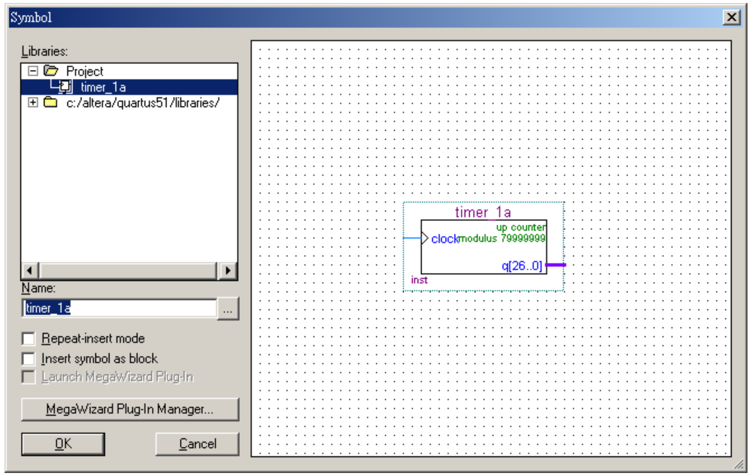


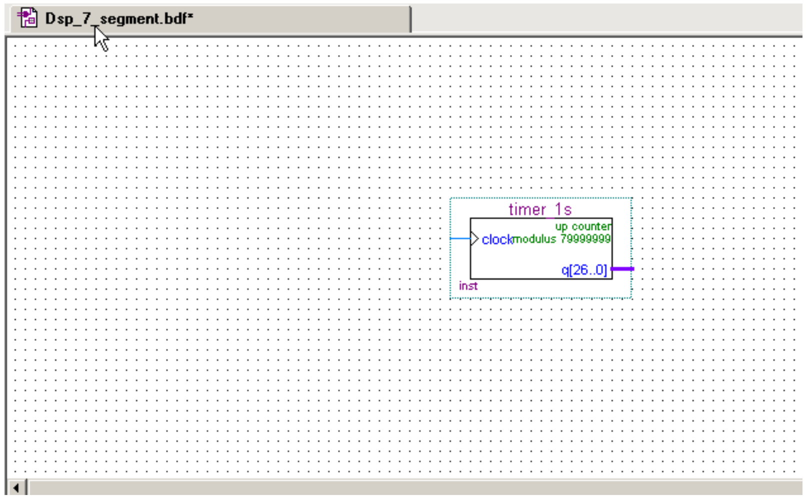




Step 4

In the **Graphic Editor**, **double-click** in the screen so that the **Symbol** Window appears. Inside the symbol window, **click** on **+** to expand the symbols defined in the **Project** folder. **Double-click** on **timer_1s. Click** the **left mouse button** to put down the symbol inside the schematic file.. *The symbol for* **"timer_1s"** *now appears in the schematic.*

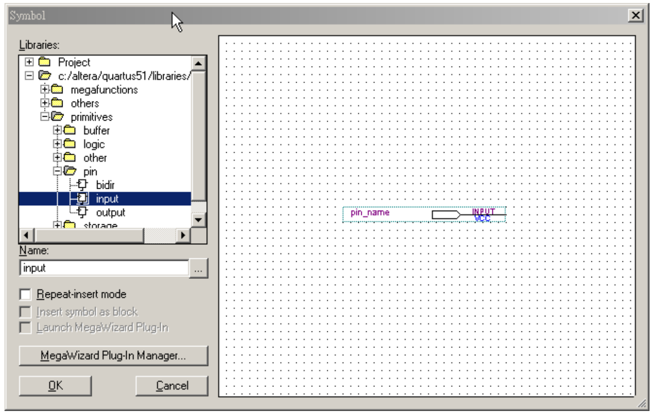




Step 5 Add Pins to the Design

Input	**Output**
sys_clk	7_out[6..0]
reset	Dig1

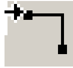
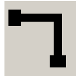
For each of the pins listed in left Table , you must insert a pin and change its name

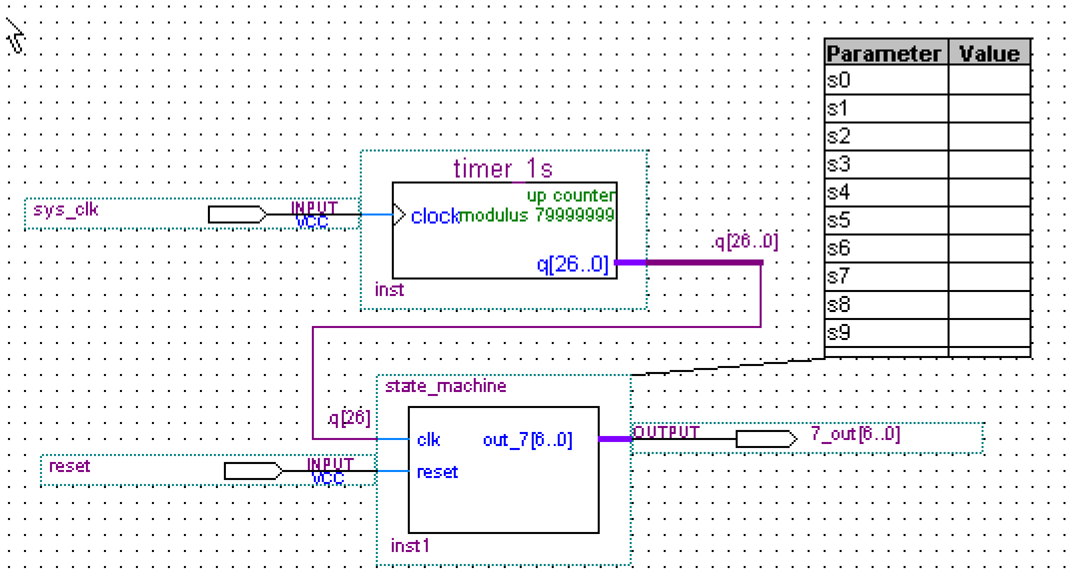


1. To place pins in the schematic file, go to **Edit** ⇒ **Insert** ⇒ **Symbol** <u>OR</u> **double-click** in any empty location of the **Graphic Editor**.
2. Browse to **libraries** ⇒ **primitives** ⇒ **pin** folder. **Double-click** on **input** or **output** *Hint: To insert multiple pins select **Repeat Insert Mode**.*
3. To rename the pins **double-click on the pin name after it has been inserted.**
4. Type the name in the Pin name(s) field and Click OK

.

Step 6 Connect the Pins and Blocks in the Schematic

1. In the left hand tool bar click on  button to draw a wire and  button to draw a bus. Another way to draw wires and busses is to place the cursor next to the port of any symbol. When you do this, the wire or bus tool will automatically appear.
2. Connect all of the pins and blocks as shown in the **figure below**



Step 7 Save and check the schematic

1. Click on the **Save** button in the toolbar  to save the schematic.
2. From the Project menu, select Add/Remove Files in Project.
   Click on the browse button to make sure the **the file**,
   timer_1s and are added to the project.
3. From the **Processing** menu, select **Start** ⇒ **Start Analysis & Elaboration**.

   Analysis and elaboration checks that all the design files are present and connections have been made correctly.
4. *Click **OK** when analysis and elaboration is completed*

Quartus II Basic Training

Quartus II Quick Start

Part3

Objectives

- *Pin assignment*
- *Perform full compilation Build a design using the schematic editor*
- *How to Download programming file*

Step 1

1. Choose **Assignments** ⇒ Assignment editor.
2. From the **View** menu, select **Show All Know Pin Names.**
3. Please click **Pin** in **Category**

	To	Location	I/O Bank	I/O Standard	General Function	Sp
1	7_out			LVTTL		
2	7_out[0]			LVTTL		
3	7_out[1]			LVTTL		
4	7_out[2]			LVTTL		
5	7_out[3]			LVTTL		
6	7_out[4]			LVTTL		
7	7_out[5]			LVTTL		
8	7_out[6]			LVTTL		
9	reset			LVTTL		
10	sys_clk			LVTTL		
11	<<new>>	<<new>>				

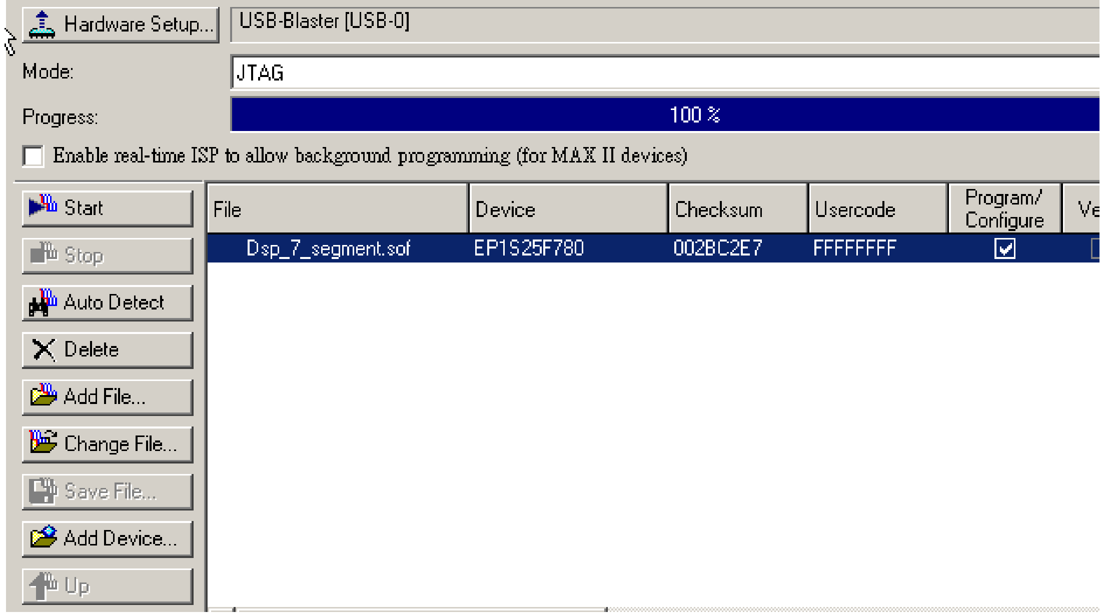
Step 2

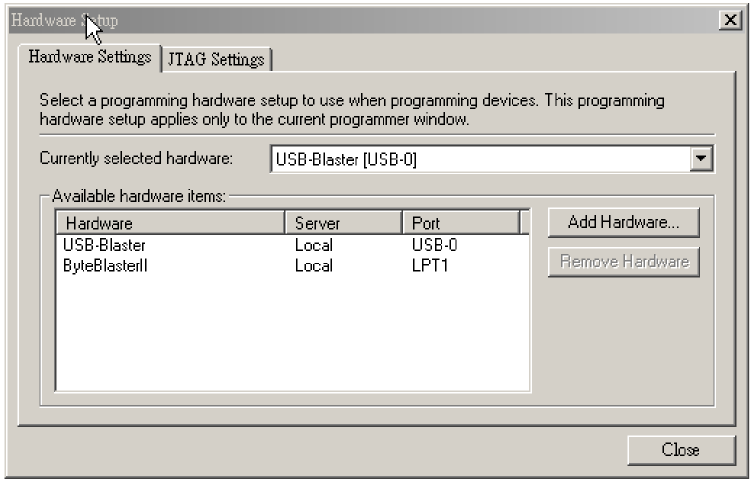
1. Pls install DSP Development Kit Stratix edtion CD
2. Open ds_stratix_dsp_bd.pdf from C:\megacore\stratix_dsp_kit-v1.1.0\Doc
3. Check clk , pushbotton and seven segment display pin location from ds_stratix_dsp_bd.pdf
4. Key your pin number in location
5. Click on the **Save** button in the toolbar
6. From **Assignments**, select **Device**. Click **Device & Pin options**. Click **Unused pins** .Select **As input tri-stated** from **Reserve all unused pins**
7. From the **Processing** menu, select **Start Compilation**

	To	Location
1	7_out[0]	PIN_L18
2	7_out[1]	PIN_D24
3	7_out[2]	PIN_L23
4	7_out[3]	PIN_L24
5	7_out[4]	PIN_L22
6	7_out[5]	PIN_L20
7	7_out[6]	PIN_L19
8	reset	PIN_F24
9	sys_clk	PIN_K17
10	7_out	

Step 3

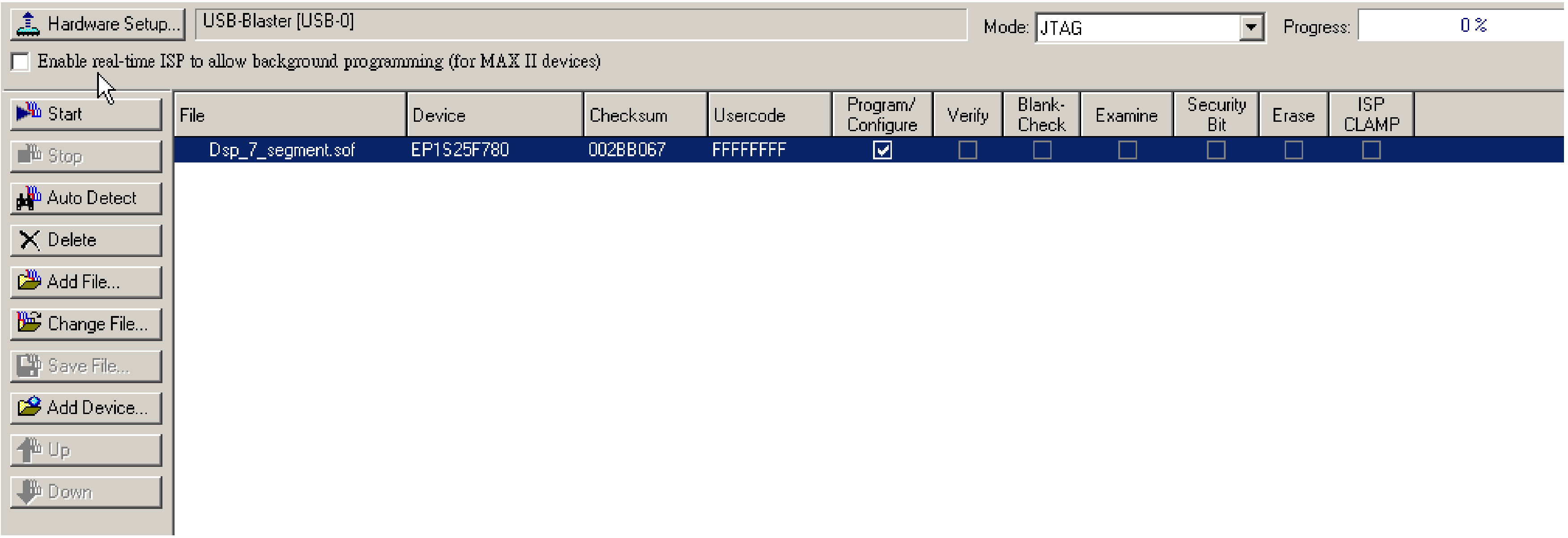
1. From the **Tools** menu, select **programmer**
2. Click on **Add File**.
3. Check **Hardware Setup.** Select your download cable **on Currently selected hardware(ByteBlasterII)**
4. Select **JTAG** from **Mode**





Step 4

1. Turn on **Program/configure. Or see figure below**
2. **Click Start**
3. **See 7-segment status**



Supported Download Cables

- USB Blaster
  - USB Port Cable
- ByteBlaster™ II
  - Parallel Port Cable
- ByteBlasterMV™
  - Parallel Port
- MasterBlaster™
  - USB / Serial Port Cable

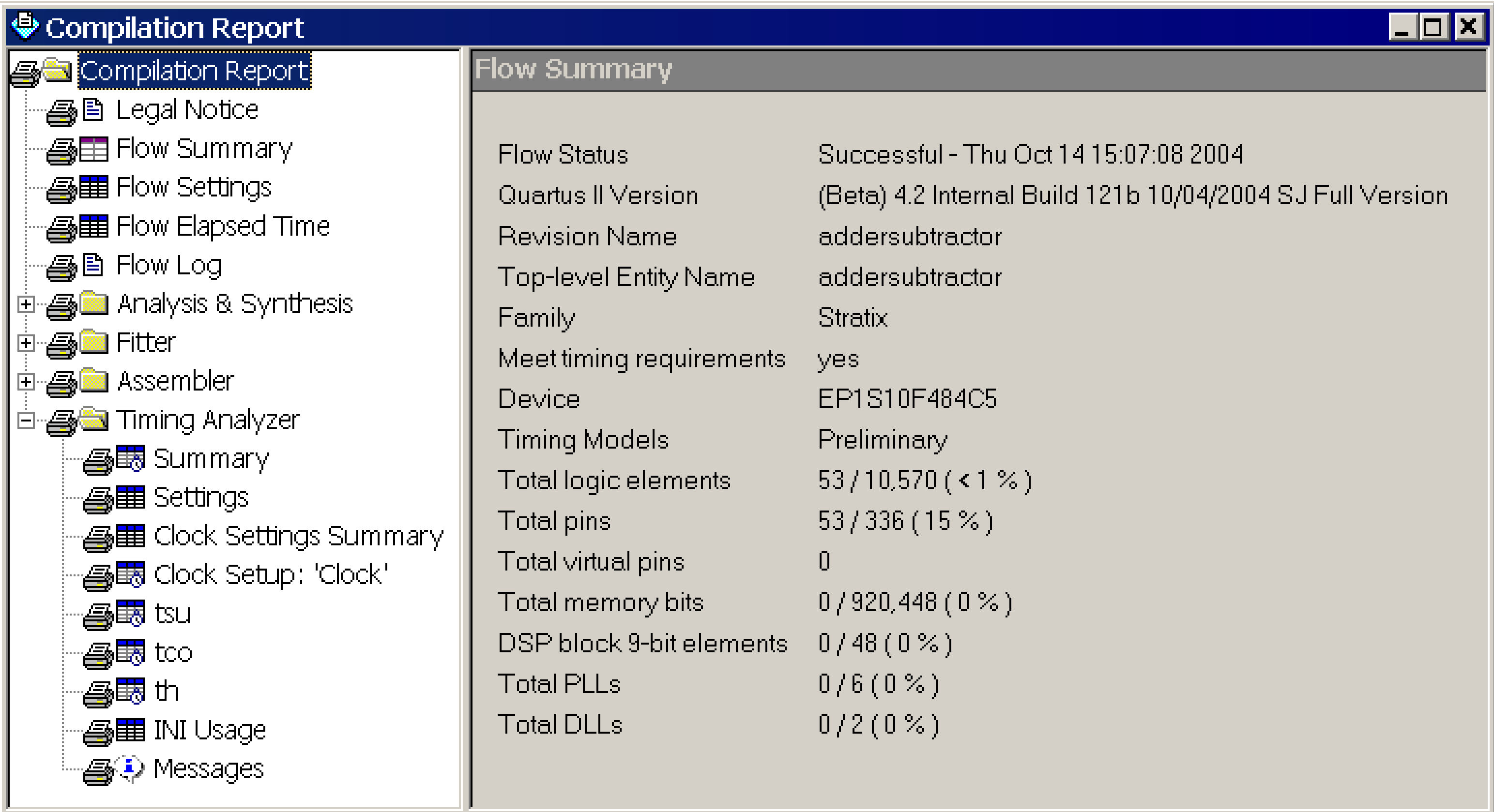
Quartus II Basic Training

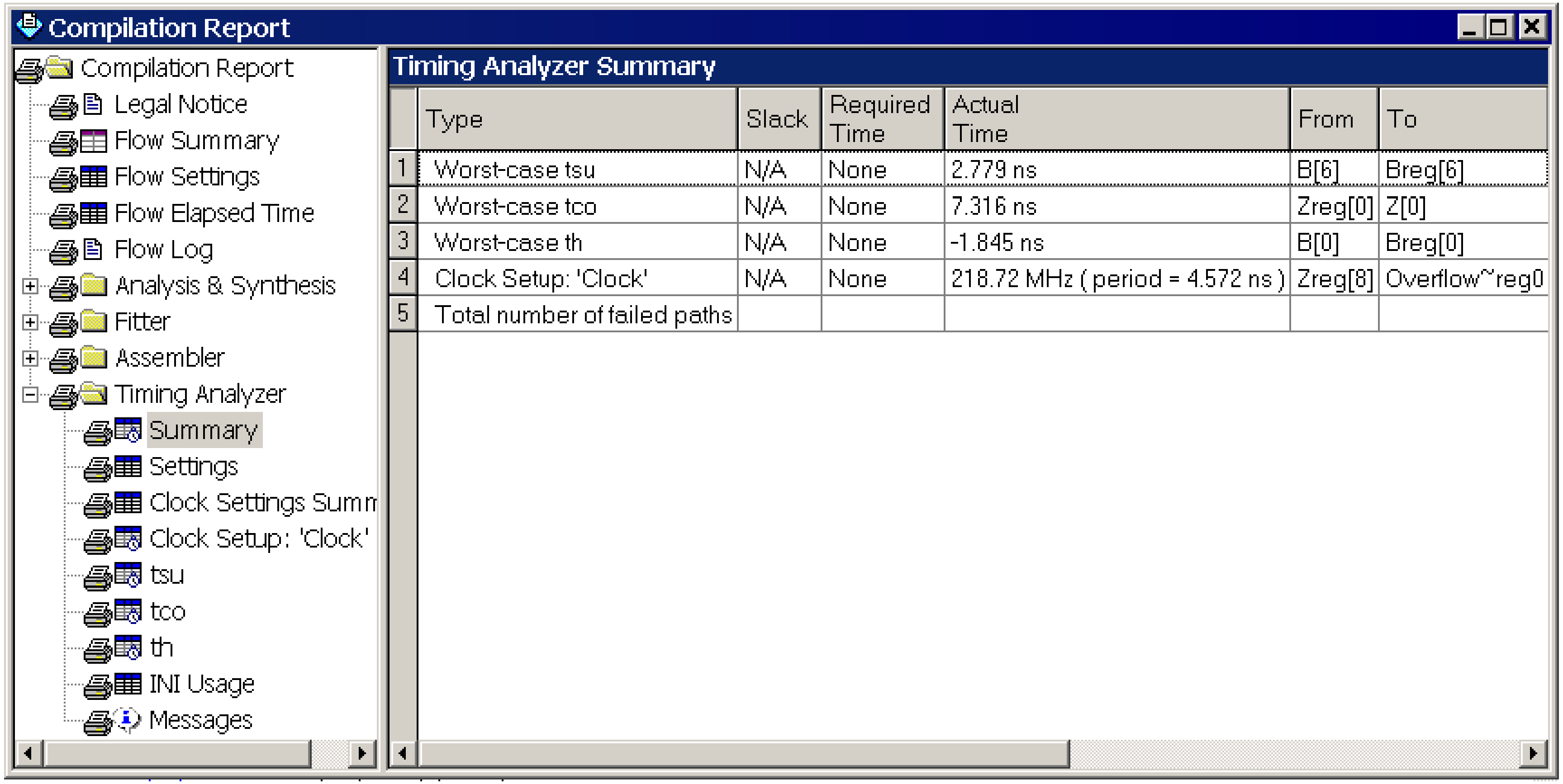
Quartus II Quick Start

Part4

Analyzing Synthesis Result

- Processing > *Compilation Report*

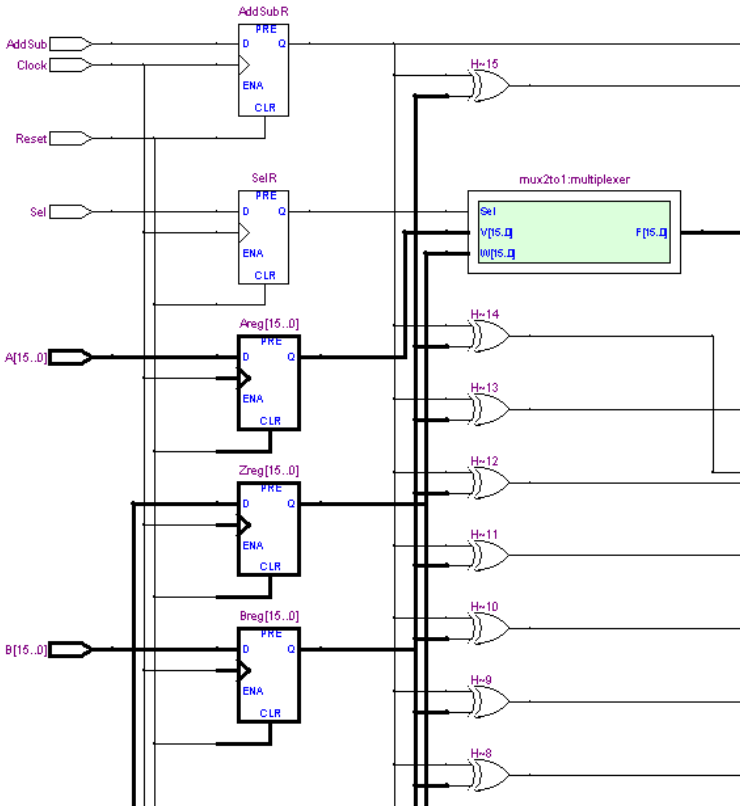




	Type	Slack	Required Time	Actual Time	From	To
1	Worst-case tsu	N/A	None	2.779 ns	B[6]	Breg[6]
2	Worst-case tco	N/A	None	7.316 ns	Zreg[0]	Z[0]
3	Worst-case th	N/A	None	-1.845 ns	B[0]	Breg[0]
4	Clock Setup: 'Clock'	N/A	None	218.72 MHz (period = 4.572 ns)	Zreg[8]	Overflow~reg0
5	Total number of failed paths					

RTL Viewer

- Tools > *RTL Viewer*



Specifying Timing Constraints

- Assignments > Timing Settings

Thank You